

V vs Go DS50 Benchmark

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Abstract

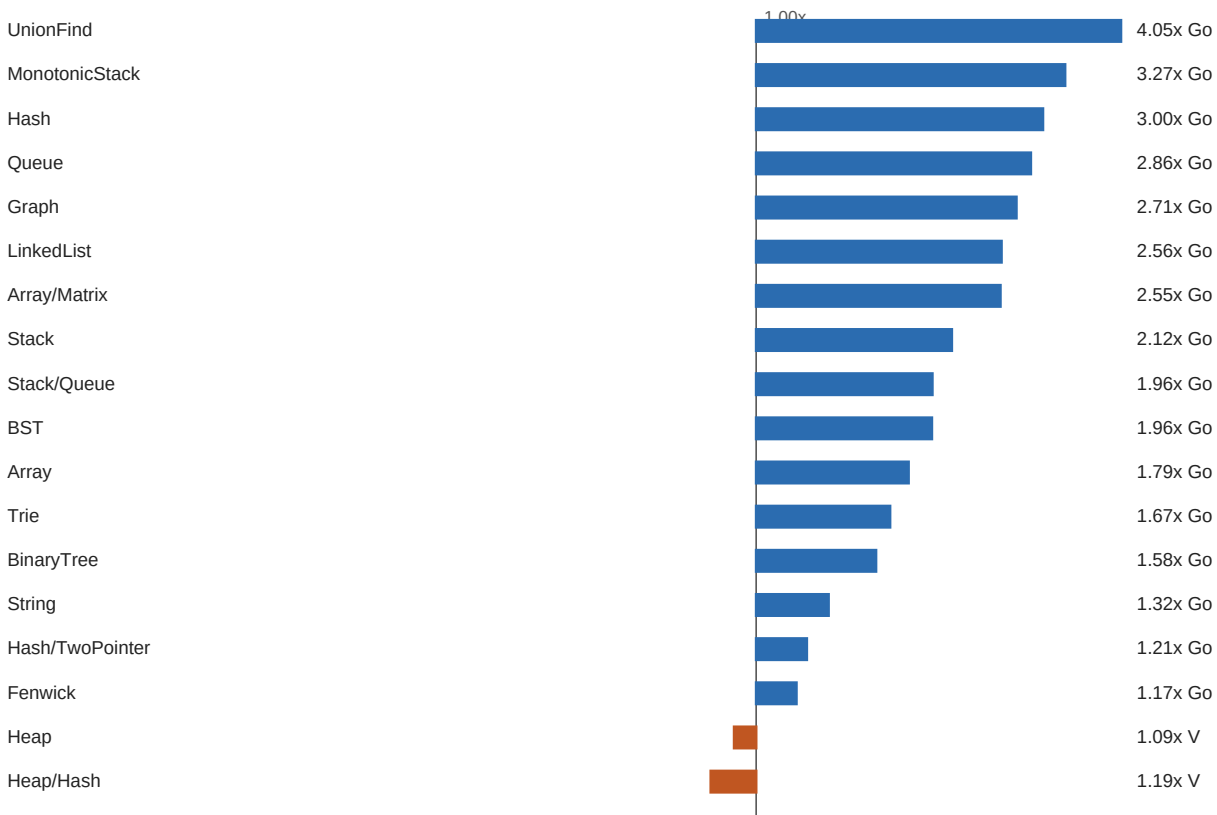
This report compares V and Go implementations of 50 classic data-structure workloads. Each workload is executed at the same logical input sizes and summarized with median elapsed time. The overall geomean result is **1.96x Go** across valid problem-size measurements.

Methodology

The benchmark uses sizes 512, 2048, 8192 with 20 measured trial(s) per language/problem/size. The Windows runner executes benchmark and compile commands at Idle priority with CPU affinity limited by the experiment script and with a Job Object memory ceiling. Checksums are compared between languages to catch behavioral drift before interpreting speed results.

Aggregate Results

Category Geomean Speedups



Category	Measurements	Geomean	Winner
Array	21	1.793	1.79x Go
Array/Matrix	3	2.549	2.55x Go
BST	6	1.960	1.96x Go
BinaryTree	24	1.583	1.58x Go
Fenwick	3	1.167	1.17x Go

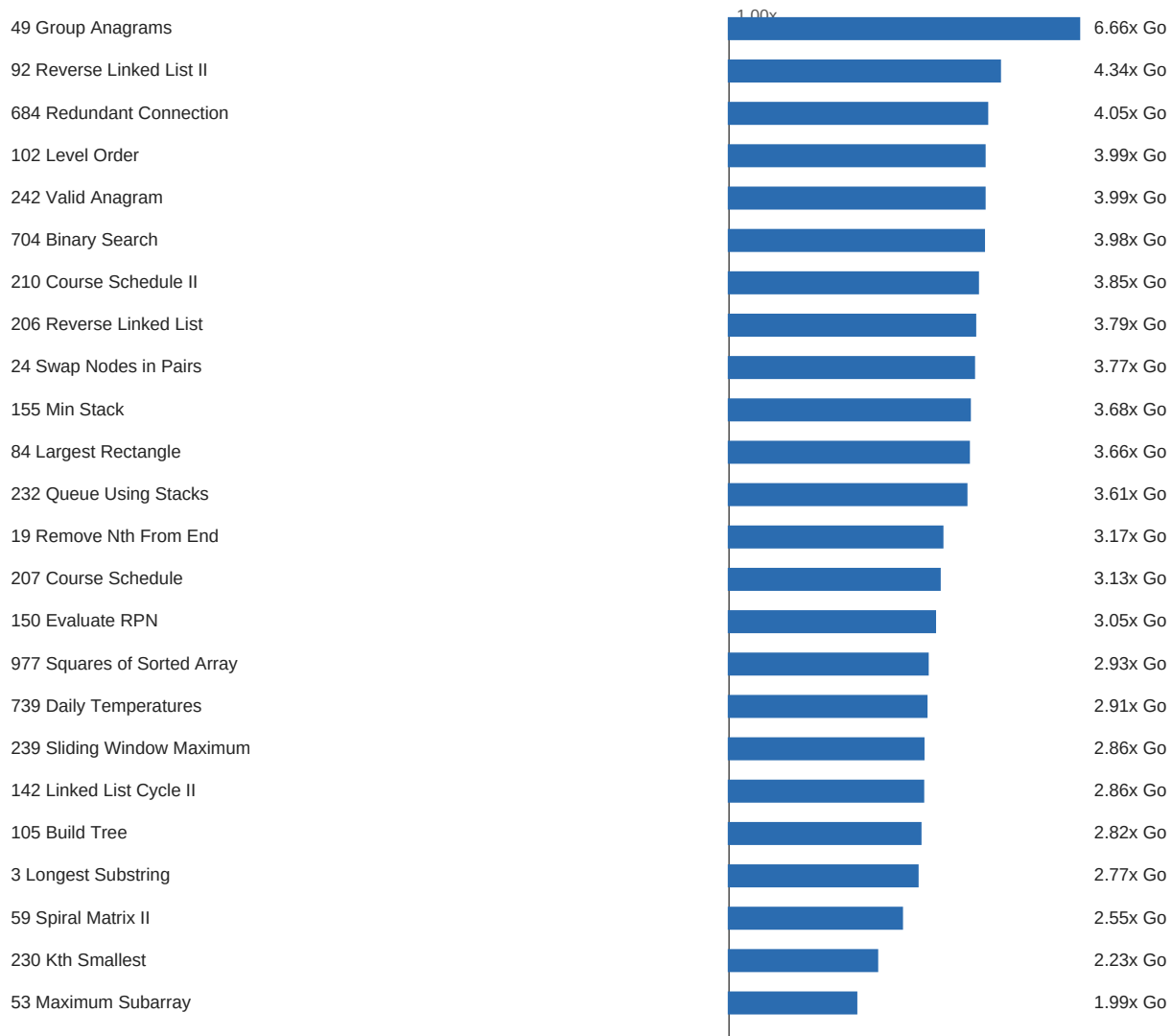
Category	Measurements	Geomean	Winner
Graph	9	2.710	2.71x Go
Hash	12	3.000	3.00x Go
Hash/TwoPointer	3	1.214	1.21x Go
Heap	6	0.919	1.09x V
Heap/Hash	3	0.840	1.19x V
LinkedList	21	2.560	2.56x Go
MonotonicStack	6	3.266	3.27x Go
Queue	3	2.864	2.86x Go
Stack	9	2.116	2.12x Go
Stack/Queue	6	1.964	1.96x Go
String	9	1.320	1.32x Go
Trie	3	1.670	1.67x Go
UnionFind	3	4.045	4.05x Go

Input Size Sensitivity

Input Size	Measurements	Geomean	Winner
512	50	2.012	2.01x Go
2048	50	1.957	1.96x Go
8192	50	1.927	1.93x Go

Problem-Level Results

Problem Geomean Speedups (Top 24 by Go Speedup)



Problem Summary Table

ID	Category	Title	Geomean	Winner
1	Array	704 Binary Search	3.975	3.98x Go
2	Array	27 Remove Element	1.896	1.90x Go
3	Array	977 Squares of Sorted Array	2.930	2.93x Go
4	Array	209 Minimum Size Subarray Sum	1.693	1.69x Go
5	Array/Matrix	59 Spiral Matrix II	2.549	2.55x Go
6	Array	53 Maximum Subarray	1.990	1.99x Go
7	Array	56 Merge Intervals	0.579	1.73x V
8	Array	238 Product Except Self	1.382	1.38x Go
9	Hash	1 Two Sum	1.796	1.80x Go
10	Hash	242 Valid Anagram	3.989	3.99x Go
11	Hash	49 Group Anagrams	6.659	6.66x Go
12	Hash	454 4Sum II	1.698	1.70x Go
13	Hash/TwoPointer	15 3Sum	1.214	1.21x Go
14	String	3 Longest Substring	2.774	2.77x Go
15	String	76 Minimum Window	1.360	1.36x Go
16	String	438 Find Anagrams	0.609	1.64x V
17	LinkedList	707 Design Linked List	1.677	1.68x Go
18	LinkedList	206 Reverse Linked List	3.790	3.79x Go
19	LinkedList	92 Reverse Linked List II	4.335	4.34x Go
20	LinkedList	24 Swap Nodes in Pairs	3.767	3.77x Go
21	LinkedList	19 Remove Nth From End	3.174	3.17x Go
22	LinkedList	160 Intersection Linked List	0.764	1.31x V
23	LinkedList	142 Linked List Cycle II	2.860	2.86x Go
24	Stack	20 Valid Parentheses	0.844	1.19x V
25	Stack/Queue	232 Queue Using Stacks	3.615	3.61x Go
26	Stack/Queue	225 Stack Using Queues	1.067	1.07x Go
27	Stack	155 Min Stack	3.682	3.68x Go
28	Stack	150 Evaluate RPN	3.049	3.05x Go
29	Queue	239 Sliding Window Maximum	2.864	2.86x Go
30	MonotonicStack	739 Daily Temperatures	2.912	2.91x Go
31	MonotonicStack	84 Largest Rectangle	3.663	3.66x Go
32	Heap	215 Kth Largest	0.990	1.01x V

ID	Category	Title	Geomean	Winner
33	Heap/Hash	347 Top K Frequent	0.840	1.19x V
34	Heap	295 Median Finder	0.853	1.17x V
35	BinaryTree	102 Level Order	3.989	3.99x Go
36	BinaryTree	104 Maximum Depth	1.370	1.37x Go
37	BinaryTree	110 Balanced Binary Tree	0.362	2.77x V
38	BinaryTree	226 Invert Binary Tree	1.542	1.54x Go
39	BinaryTree	101 Symmetric Tree	1.915	1.91x Go
40	BinaryTree	112 Path Sum	1.934	1.93x Go
41	BinaryTree	105 Build Tree	2.818	2.82x Go
42	BinaryTree	236 Lowest Common Ancestor	1.238	1.24x Go
43	BST	98 Validate BST	1.725	1.72x Go
44	BST	230 Kth Smallest	2.228	2.23x Go
45	Graph	200 Number of Islands	1.654	1.65x Go
46	Graph	207 Course Schedule	3.127	3.13x Go
47	Graph	210 Course Schedule II	3.848	3.85x Go
48	UnionFind	684 Redundant Connection	4.045	4.05x Go
49	Trie	208 Implement Trie	1.670	1.67x Go
50	Fenwick	307 Range Sum Query Mutable	1.167	1.17x Go

Validity Checks

Raw records: 6000. Compared rows: 150. Checksum mismatches: 0. Go wins: 126; V wins: 23; ties: 1.